Directional Navigability Estimation from Ultra-Low-Resolution Multi-View Thermal Sensing

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***Abstract*—Reliable perception of navigable indoor spaces remains a challenge for wearable navigation systems, which typically rely on high-resolution cameras, depth sensors, or large annotated datasets—limiting deployability in privacy-sensitive environments and raising development costs. This paper investigates whether directional navigability can be estimated using only ultra-low-resolution thermal imagery from a wearable multi-camera platform, a setting made difficult by the limited detail of low-resolution thermal sensors and the cost of annotating thermal data. We propose a calibration-free learning framework that uses high-resolution RGB imagery during data collection to automatically generate navigability labels, enabling training without manual annotation. The resulting models estimate multi-direction navigability directly from multi-view thermal observations, requiring no geometric calibration, pixel-level alignment, or fixed sensor configuration. Experiments with a wearable rig of three ultra-low-resolution thermal cameras show that directional navigability can be reliably inferred from thermal sensing alone. We further compare multi-camera fusion strategies and evaluate robustness to camera-arrangement changes, highlighting ultra-low-resolution thermal sensing as a scalable, privacy-preserving solution for wearable indoor navigation.**

I. Introduction

Wearable navigation aids promise greater independence for visually impaired users in unfamiliar indoor spaces, but their sensing stacks are costly. Most rely on high-resolution RGB or active depth sensors, which continuously capture identifiable imagery of bystanders and raise privacy concerns when worn indoors. They also require large hand-labeled datasets to learn which directions are walkable, and typically assume a fixed, geometrically calibrated camera layout that breaks as soon as a sensor shifts.

Ultra-low-resolution thermal sensing is an appealing alternative. A 32×24-pixel array costs a few tens of dollars, draws little power, is invariant to lighting, and is too coarse to resolve identifiable facial features. These same properties make perception hard: each frame carries little spatial detail, several cameras are needed for useful coverage, and annotating thermal data is laborious. We therefore ask whether *directional navigability*—which directions around the wearer are walkable—can be estimated from ultra-low-resolution thermal imagery alone, without geometric calibration and without manual labels.

We answer this affirmatively with a calibration-free, RGB-supervised framework (Fig. 1). During data collection only, a high-resolution RGB camera rides alongside a wearable rig of three ultra-low-resolution thermal cameras; an off-the-shelf detector run on the RGB frames automatically generates per-direction navigability labels, so no manual annotation is needed. At inference the RGB camera is removed: the model consumes only the three raw thermal tiles and predicts, for each of the left, middle, and right directions, whether the space is clear, occupied by a person, or blocked by another object. Because supervision is transferred through labels rather than pixel-aligned feature matching, the framework needs no calibration, no pixel-level alignment, and no fixed correspondence between sensor position and input channel.

Within this framework we study two deployment-critical questions—how to fuse the three views, and how robust the system is to camera placement—which motivate our contributions: (i) a calibration-free framework that transfers navigability supervision from an RGB detector to a thermal-only model, with no labels or RGB at deployment; (ii) a direct comparison of three multi-view fusion strategies (pixel-, decision-, and feature-level) on ultra-low-resolution thermal data; and (iii) an analysis showing that a lightweight camera-identity correction restores accuracy when camera slots are shuffled, matching a ground-truth oracle re-sort.

II. Related Work

A. Privacy-Preserving and Ultra-Low-Resolution Thermal Sensing

Wearable indoor navigation systems for visually impaired users have traditionally relied on RGB cameras or depth sensors, which raises privacy concerns when deployed continuously indoors. Gao *et al.*'s DareDevil [1] addresses this by auto-labeling thermal frames from a paired RGB camera through a pretrained YOLOv8 detector, avoiding manual annotation. However, DareDevil collapses all object classes into a single binary obstacle label and operates on colorized thermal frames from a single standard-resolution thermal camera at inference time; it does not address the extreme spatial compression of a 32×24-pixel array, multi-camera fusion, or robustness to sensor placement.

A separate line of work uses ultra-low-resolution thermal sensors (e.g., 24×32 arrays, on the order of $30 per unit) as a low-power, lighting-invariant alternative to RGB and depth sensing. Such sensors are too coarse to resolve identifiable facial features, but their reduced spatial detail also degrades the features downstream perception models rely on; prior work compensates for this by fusing thermal imagery with gyroscope readings for motion estimation [5]. Naser *et al.* [4] separately showed that most low-resolution thermal pipelines

[PLACEHOLDER]

Full-width pipeline figure.

End-to-end overview: wearable three-camera ultra-low-resolution thermal capture (RGB camera used only at data-collection time) → RGB/YOLO automatic per-direction label generation → the four thermal fusion models (stitched panorama, output fusion, feature fusion, feature fusion with camera-identity correction) → per-direction navigability predictions.

Fig. 1. End-to-end pipeline, from multi-view thermal data collection and RGB-based automatic labeling to the family of thermal fusion models evaluated in this paper. The RGB camera is present only during data collection; at inference the models use the three ultra-low-resolution thermal tiles alone.

still assume a fixed, calibrated sensor geometry. This work targets discrete navigability classification rather than continuous odometry, and removes the dependence on geometric calibration that these systems retain.

B. Automatic Label Generation for Traversability

Manually annotating walkable regions is expensive and does not scale to real-world indoor environments. Kim *et al.* [2] addressed this for egocentric video with an automated annotation strategy that derives traversability labels without manual pixel-level annotation. We apply this same philosophy to a cross-modal setting: RGB-derived object detections supervise a thermal-only model only during data collection, and the RGB sensor is entirely absent at deployment.

C. Large-Scale and Web-Sourced Navigation Learning

CityWalker [3] trains an embodied urban navigation policy on over 2,000 hours of in-the-wild web video, using visual odometry to generate pseudo-labels instead of teleoperated demonstrations. CityWalker targets continuous waypoint-following from RGB video; this work targets discrete per-direction navigability from privacy-preserving thermal sensing, with an order of magnitude less data collected from a purpose-built wearable rig.

D. Multi-View Sensor Fusion and Placement Robustness

Multi-camera perception systems typically assume a fixed mapping between physical sensor position and input channel, an assumption that breaks when sensors shift or are reattached in a different order on a wearable rig. Pixel-level, decision-level, and feature-level fusion are well studied in the sensor fusion literature, but little work examines how these strategies behave when the camera-to-slot assignment is perturbed, particularly at the small camera counts and extreme resolutions considered here. This also relates to cross-resolution knowledge transfer, where a high-resolution signal supervises a model on coarser inputs [6]; our gap is larger still, since the supervisory RGB detector and the deployed 32×24 thermal arrays share no feature space and no geometric alignment at inference time. We contribute a direct comparison of three fusion strategies for an ultra-low-resolution, three-camera array, together with an explicit treatment of camera-slot permutation robustness.

III. Methodology

A. Problem Formulation

At each timestep the wearable rig produces three ultra-low-resolution thermal tiles $\{x_{\mathrm{L}}, x_{\mathrm{M}}, x_{\mathrm{R}}\}$, one per active camera, each of size 32×24. The task is to predict, for every direction $d \in \{\text{left}, \text{middle}, \text{right}\}$, a navigability state $y_d \in \{\texttt{no_object}, \texttt{person}, \texttt{other_object}\}$, where `no_object` denotes a clear (walkable) direction and the remaining two states denote an obstruction, separated by whether the obstruction is a person or another object. A model is thus a map $f : (x_{\mathrm{L}}, x_{\mathrm{M}}, x_{\mathrm{R}}) \mapsto (y_{\mathrm{L}}, y_{\mathrm{M}}, y_{\mathrm{R}})$ that operates on the raw thermal tiles alone at inference time. Supervision comes from a paired high-resolution RGB camera used only during data collection (Section III-B), and each direction carries a validity mask so that timesteps without a usable RGB label do not contribute to the loss.

B. RGB-Based Automatic Label Generation

Ground-truth navigability labels are generated automatically from the paired RGB stream, eliminating manual annotation. For each synchronized RGB frame we run a pretrained YOLOv8 detector; detections of the COCO `person` class are mapped to the `person` state and all other detected object classes to `other_object`. To assign detections to directions, each detection's image-space bounding box is tested for overlap against the horizontal footprint of each active thermal camera's field of view, recovered from the RGB-to-thermal homographies estimated once at rig-build time. A direction is labeled `person` if a person detection overlaps its footprint, `other_object` if only a non-person detection overlaps, and `no_object` (clear) otherwise. Frames without a reliable homography-derived footprint are marked invalid for that direction and masked out of training. This yields per-direction, three-class labels aligned to the thermal tiles without any pixel-level correspondence between the RGB and thermal sensors at inference time.

C. Thermal Fusion Architectures

We evaluate four architectures for estimating per-direction navigability from the three active thermal cameras (left, middle, right), each consuming only raw 32×24 thermal tiles at inference time and producing a 3-class prediction per camera over $\{\texttt{no_object}, \texttt{person}, \texttt{other_object}\}$.

The four span pixel-level, decision-level, and feature-level fusion, plus a fourth variant that adds a learned correction for camera-placement permutations on top of feature-level fusion. All but the stitched-panorama baseline share a configurable convolutional backbone, `FlexBackbone` (depth $d \in \{2, \ldots, 5\}$, width multiplier w, activation $a \in \{\text{ReLU}, \text{GELU}, \text{LeakyReLU}\}$), which returns the intermediate feature map at every block so downstream heads can pool over multiple resolutions.

a) Model 1: Stitched Panorama (pixel-level fusion): The three raw tiles are warped into a shared panoramic canvas via pre-computed thermal-to-thermal homographies, producing a two-channel panorama (raw intensity plus a validity mask). A fixed five-layer backbone is applied independently to each camera's bounding-box crop of the panorama, followed by a shared classification head. Because the panorama is built once, before any learned processing, cross-camera information lost to the homography warp is unavailable to the model; this baseline isolates the cost of fusing at the pixel level.

b) Model 2: Output Fusion (decision-level fusion): A shared-weight `FlexBackbone` processes each camera's tile independently, and a per-camera head reduces it to one 3-class logit vector. The three logit vectors are concatenated and passed through a small fusion MLP that refines them jointly. Fusion only ever sees already-classified, low-dimensional logits, so cross-camera interaction is limited to statistical dependencies between directions rather than raw spatial features.

c) Model 3: Feature Fusion: The same shared-weight backbone runs on each camera's tile, but instead of classifying each camera separately, the feature maps from all three cameras at a given backbone level are concatenated channel-wise and projected back to the original channel count via a learned 1×1 convolution, applied at every level. This yields a single fused multi-scale feature pyramid mixing all three cameras at every depth, in contrast to Model 2's late fusion. A shared head then produces the final per-camera logits. We sweep backbone depth, width, activation, and head capacity to select the configuration used as the reference Feature Fusion result throughout this paper.

d) Model 4: Feature Fusion with camera-identity correction: Model 3's per-level 1×1 projections consume all cameras' features in a fixed slot order, so the projection layer can learn to associate an input slot with a physical direction even though the backbone itself is fully shared and permutation-agnostic; if a sensor is reattached in a different slot at deployment time, this learned positional dependence can degrade predictions. We correct for it with a lightweight auxiliary `CameraIdentityClassifier`, a compact CNN that takes one raw thermal tile and predicts a 3-way softmax over $\{\text{left}, \text{middle}, \text{right}\}$, trained independently using the true camera-to-direction mapping recorded at capture time. At inference, given a possibly shuffled set of tiles, the classifier scores every slot against every direction and picks the assignment that maximizes total log-probability, enumerated exhaustively over all $3! = 6$ permutations; the resorted tiles are then passed to the unmodified Model 3. We compare this against an oracle re-sort using ground-truth identities (an upper bound on achievable robustness) and a blind condition with no correction at all.

All four architectures share the same masked, per-camera, three-class cross-entropy training objective, gated by a per-camera validity mask so that timesteps without a labeled RGB frame do not contribute gradient.

IV. Experimental Setup

A. Wearable Thermal Data Collection Platform

We developed a wearable thermal sensing platform designed for privacy-preserving indoor navigation research. The platform consists of multiple ultra-low-resolution thermal cameras mounted on a wearable frame and connected to an embedded computing unit for synchronized data acquisition. The design emphasizes low cost, privacy preservation, and flexibility, making it suitable for studying navigation perception under severe sensing constraints.

The wearable platform incorporates multiple MLX90640 SparkFun IR Array Breakout sensors, each providing a 32×24 thermal image of the environment. The sensors are mounted at different viewing angles to increase environmental coverage and capture complementary observations of the user's surroundings. All sensors are connected to an NVIDIA Jetson Nano-based acquisition unit that records timestamped thermal frames for subsequent processing.

B. Data Collection

Data were recorded by walking the wearable rig through indoor environments while a high-resolution RGB camera, rigidly mounted alongside the thermal array, captured a time-synchronized reference stream used only for label generation. During each traversal the three active thermal cameras and the RGB camera recorded timestamped frames continuously, producing sequences that span clear corridors as well as encounters with people and static objects, so that all three navigability states are represented across the left, middle, and right directions. Only the thermal tiles and their automatically generated per-direction labels (Section III-B) are retained for model training; the RGB stream is discarded after labeling and is never available at inference.

C. Training Protocol

To avoid leaking temporally adjacent, near-duplicate frames across splits, we use a blocked partition: each recording is divided into contiguous blocks and the final 20% of every block is held out for validation, giving a train/validation split that keeps correlated neighbors together. Camera-identity-shuffle experiments are additionally evaluated on a held-out test set. All learned models share the same masked, per-camera, three-class cross-entropy objective and are trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}) under a cosine-annealing schedule for 30 epochs at batch size 64, with class weighting to counter the imbalance toward clear directions. For Feature Fusion we additionally sweep backbone depth, width, activation, and head capacity and report the best

[PLACEHOLDER]

Bar chart 1: architecture comparison.

Per-class F_1 by camera direction for the four architectures (`fig1_macro_f1_color.png`).

Fig. 2. Per-class F_1 by camera direction for the obstruction-detection architectures. Baselines and Feature Fusion are evaluated on the validation set; Feature Fusion with shuffle correction (blind, classifier, oracle) is evaluated on the held-out test set.

configuration. We evaluate with per-class F_1 score reported per direction (left, middle, right) and overall; because the label distribution is skewed toward clear directions, macro-averaged F_1 is used for model selection rather than raw accuracy.

V. Results

A. Navigability Estimation Performance

Figure 2 compares the four architectures using per-class F_1 broken down by camera direction. All models estimate the three navigability states well above chance from the raw 32×24 thermal tiles, confirming that directional navigability is recoverable from ultra-low-resolution thermal sensing alone. On the validation set, Feature Fusion is the strongest of the calibrated (non-shuffled) models, reaching overall per-class F_1 of roughly 0.81 (clear), 0.83 (person), and 0.80 (other object). The stitched-panorama and output-fusion baselines trail it, with overall F_1 near 0.73–0.78, and both show markedly weaker person detection on the right camera, where the useful thermal signal is sparsest.

B. Fusion Strategy Analysis

Feature-level fusion consistently outperforms both pixel-level stitching and decision-level output fusion. Mixing intermediate feature maps across cameras at every backbone depth preserves cross-camera spatial detail that the alternatives discard: the stitched panorama warps and blurs the already-coarse thermal signal before the model sees it, while output fusion only combines low-dimensional, already-classified logits. Feature fusion costs extra backbone passes, but on these ultra-low-resolution tiles the accuracy gain justifies the compute, so we adopt it as the reference architecture for the robustness study.

[PLACEHOLDER]

Bar chart 2: shuffle-corrected Feature Fusion.

Per-class F_1 by direction and overall (`fig2_ff_per_class_color.png`).

Fig. 3. Per-class F_1 of Feature Fusion with classifier-based shuffle correction on the held-out test set, broken down by camera direction (left, middle, right) and overall.

C. Camera Arrangement Robustness

Figure 3 reports Feature Fusion under permuted camera slots on the held-out test set. Without correction (blind), shuffling the physical camera-to-slot assignment collapses performance to overall per-class F_1 near 0.40–0.42, confirming that the fixed-slot projection layers learn a positional dependence that does not survive re-arrangement. The lightweight camera-identity classifier, which re-sorts tiles to their most likely directions before the unmodified model, restores overall F_1 to roughly 0.90–0.92 across all three classes—matching the ground-truth oracle re-sort to within rounding. Once tiles are routed to the correct directions the underlying features are themselves directionally meaningful, so the system is robust to camera placement rather than overfit to which camera feeds which input channel.

Acknowledgment

The authors would like to thank...

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